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The equation $2x^4 + x + y^3 - y + 2 = 0$ has no integer solutions

Size $h = 46$, length $l = 11$

Theorem 1. *The equation*

$$2x^4 + x + y^3 - y + 2 = 0 \tag{1}$$

has no integer solutions.

The proof combines quadratic Hilbert symbols over $\mathbb{Q}$ and cubic Hilbert symbols over $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. To a hypothetical integer solution we attach the values of auxiliary polynomials H_i with rational coefficients, linked by a graph, and the values of auxiliary polynomials J_i with coefficients in $\mathbb{Z}[\omega]$, linked by a weighted graph. By the product formula and the reciprocity law, the products of the quadratic symbols over all places are 1, and the sums of the cubic symbols over all places are 0. We show that the local contributions vanish at the primes outside a finite set S , determine the sign contributions at the real place, compute the local contributions jointly at the primes of S , and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, finite computations with residues, and basic properties of the Legendre symbol and of the quadratic Hilbert symbol, the proof uses two standard facts from class field theory: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues $1, 3, 5, 7$ modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod{3}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod{3}$, the polynomial $t^2 + t + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and ω reduces to r . The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B \rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [3, Chapter V, §3]. We write $(a, b)_v = \omega^{\theta_v(a, b)}$ with $\theta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [3, Chapter V, §3 and Chapter VI, §8].

- (H1) $\theta_v(a, bc) = \theta_v(a, b) + \theta_v(a, c)$, $\theta_v(a, b) = -\theta_v(b, a)$, and $\theta_v(a, c^3) = 0$. In particular, $\theta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 3}$.
- (H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\theta_v(a, b) = e\chi(w) - f\chi(u)$. In particular, $\theta_v(a, b) = 0$ if a and b are both units.
- (H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\theta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \theta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\theta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values θ_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $\text{cl}_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), cl_v is additive, $\text{cl}_v(a)$ determines the class of a modulo cubes, and $\theta_v(a, b) = e\chi(w) - f\chi(u)$ for $\text{cl}_v(a) = (e, \chi(u))$ and $\text{cl}_v(b) = (f, \chi(w))$. We put $\text{cl}_p(a) = \text{cl}_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod{3}$, and $\text{cl}_p(a) = (\text{cl}_{v_r}(a), \text{cl}_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod{3}$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, θ_p is the sum of the forms θ_v over the places v above p . For $p \equiv 2 \pmod{3}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 8. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $\text{cl}_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$. Then cl_3 is additive, and $\theta_v(A, B) = \text{cl}_3(A)^\top M_3 \text{cl}_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute θ_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), θ_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

We write θ_3 for the form of Lemma 8. Since $3 = -\omega^2 \pi^2$ and $4 = 2^2$, we have $\text{cl}_3(3) = (2, 2, 0, 0)$ and $\text{cl}_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 9. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $\text{cl}_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

- (i) $p \equiv 2 \pmod{3}$. If $\zeta \neq 0$, then $b = \text{cl}_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{3}$. For each root r , let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $\text{cl}_{v_r}(z)$ are those of $\text{cl}_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = \text{cl}_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.

(iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = \text{cl}_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j \text{cl}_3(3)$ and D is spanned by $\text{cl}_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $\text{cl}_3(3)$ and $\text{cl}_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 8, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m . (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. $\square$

Lemma 10. *Let p be a prime, and consider vertices i with vectors b_i , subspaces D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\theta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\theta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then*

$$\sum_{(i,j)} w_{ij} \theta_p(c_i, c_j) + \sum_i \theta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \theta_p(b_i, b_j) + \sum_i \theta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As θ_p is bilinear and $\theta_p(u, v) = -\theta_p(v, u)$, the left side equals the right side plus $\sum_i \theta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \theta_p(d_i, d_j)$, and both sums vanish. $\square$

3 The quadratic certificate

Put $F(x, y) = 2x^4 + x + y^3 - y + 2$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -2x - 2y + 1, \\ H_1 &= -3. \end{aligned}$$

The edges are $\{0, 1\}$. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -108x^8 - 108x^5 - 216x^4 - 27x^2 - 108x - 104.$$

For $i \in I = \{0\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

4 The cubic certificate

The certificate consists of 63 polynomials J_i with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i \in \mathbb{Z}[\omega]$, listed in Table 4, and of 293 edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$, listed below. For an edge (i, j) we put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij} \bmod 3$, and $\rho_{ij} = 0$ if i and j are not joined by an edge. The edges $(i, j; w_{ij})$ are (0, 12; 1), (0, 15; 2), (0, 16; 2), (0, 19; 1), (0, 26; 1), (0, 28; 1), (0, 35; 1), (0, 39; 2), (1, 2; 2), (1, 6; 2), (1, 13; 1), (1, 17; 1), (1, 22; 2), (1, 24; 2), (1, 27; 1), (1, 28; 2), (1, 33; 2), (1, 39; 2), (1, 60; 2), (2, 11; 2), (2, 18; 1), (2, 21; 1), (2, 22; 2), (2, 23; 2), (2, 32; 1), (2, 34; 1), (2, 40; 1), (2, 51; 1), (3, 5; 1), (3, 9; 1), (3, 11; 2), (3, 13; 1), (3, 17; 2), (3, 26; 1), (3, 33; 1), (3, 37; 1), (3, 44; 2), (3, 46; 1), (3, 58; 2), (4, 6; 1), (4, 9; 2), (4, 19; 2), (4, 25; 2), (4, 31; 1), (4, 35; 1), (4, 38; 1), (4, 39; 2), (4, 52; 2), (5, 8; 1), (5, 15; 2), (5, 17; 2), (5, 22; 2), (5, 27; 1), (5, 28; 1), (5, 33; 1), (5, 37; 1), (6, 7; 1), (6, 14; 2), (6, 22; 2), (6, 24; 2), (6, 28; 1), (6, 32; 1), (6, 42; 2), (6, 52; 1), (7, 12; 1), (7, 25; 2), (7, 40; 1), (7, 52; 1), (7, 53; 2), (7, 61; 2), (8, 14; 1), (8, 17; 1), (8, 18; 2), (8, 22; 2), (8, 25; 1), (8, 33; 1), (8, 36; 2), (8, 38; 2), (8, 40; 1), (8, 48; 1), (9, 11; 2), (9, 13; 2), (9, 20; 2), (9, 38; 2), (9, 55; 1), (9, 58; 1), (10, 13; 2), (10, 29; 1), (10, 33; 1), (10, 37; 1), (10, 55; 2), (10, 56; 1), (10, 59; 2), (11, 23; 1), (11, 28; 2), (11, 29; 2), (11, 43; 2), (11, 49; 1), (11, 50; 2), (11, 59; 2), (12, 19; 1), (12, 25; 1), (12, 29; 1), (12, 49; 2), (13, 29; 1), (13, 31; 1), (13, 35; 2), (13, 43; 2), (13, 44; 1), (13, 46; 2), (13, 55; 2), (13, 61; 1), (14, 16; 1), (14, 17; 1), (14, 20; 1), (14, 24; 1), (14, 27; 1), (14, 32; 1), (14, 36; 2), (14, 42; 2), (14, 43; 2), (14, 45; 2), (14, 47; 2), (14, 49; 1), (14, 53; 1), (14, 55; 1), (15, 17; 2), (15, 19; 2), (15, 24; 2), (15, 25; 1), (15, 29; 2), (15, 39; 2), (15, 42; 2), (15, 45; 1), (15, 53; 2), (16, 21; 2), (16, 34; 2), (16, 35; 2), (16, 42; 2), (16, 44; 2), (16, 50; 1), (17, 30; 1), (17, 33; 2), (17, 40; 2), (17, 41; 2), (17, 45; 1), (17, 56; 2), (18, 23; 1), (18, 26; 2), (18, 33; 2), (18, 34; 1), (18, 35; 2), (18, 40; 2), (18, 44; 1), (18, 50; 2), (18, 54; 2), (18, 58; 2), (18, 61; 1), (18, 62; 1), (19, 26; 1), (19, 33; 1), (19, 37; 1), (19, 38; 1), (19, 41; 2), (20, 23; 1), (20, 38; 2), (20, 48; 1), (20, 52; 2), (21, 23; 1), (21, 25; 1), (21, 26; 1), (21, 33; 1), (21, 35; 1), (21, 50; 2), (22, 23; 2), (22, 37; 2), (22, 51; 2), (22, 54; 2), (23, 27; 2), (23, 30; 1), (23, 41; 1), (23, 52; 1), (23, 53; 1), (24, 27; 2), (24, 33; 2), (24, 36; 2), (24, 37; 2), (24, 38; 1), (24, 39; 1), (24, 44; 1), (24, 45; 1), (24, 49; 1), (24, 53; 2), (24, 57; 2), (25, 31; 1), (25, 37; 2), (25, 39; 1), (25, 49; 1), (25, 52; 1), (25, 53; 2), (26, 27; 2), (26, 33; 2), (26, 34; 1), (26, 53; 2), (27, 40; 2), (28, 37; 2), (28, 39; 1), (29, 30; 1), (29, 39; 2), (29, 41; 2), (29, 42; 1), (29, 51; 2), (29, 54; 1), (29, 56; 2), (29, 57; 2), (29, 59; 1), (29, 60; 1), (29, 61; 1), (30, 35; 2), (30, 37; 2), (30, 41; 1), (30, 47; 1), (30, 48; 1), (30, 50; 2), (30, 52; 1), (30, 55; 1), (31, 32; 2), (31, 46; 2), (32, 39; 1), (33, 34; 1), (33, 36; 2), (33, 39; 2), (33, 46; 1), (33, 55; 1), (33, 56; 2), (33, 57; 2), (33, 59; 2), (33, 60; 1), (33, 62; 2), (34, 37; 2), (34, 40; 2), (34, 56; 1), (34, 57; 1), (35, 50; 1), (35, 52; 2), (35, 55; 2), (36, 44; 1), (36, 52; 2), (36, 57; 1), (37, 42; 1), (37, 44; 2), (37, 53; 1), (38, 42; 1), (38, 52; 1), (39, 52; 2), (39, 55; 2), (39, 56; 1), (40, 53; 1), (41, 51; 2), (41, 59; 1), (41, 61; 1), (42, 45; 1), (42, 48; 1), (42, 50; 1), (42, 53; 1), (42, 61; 1), (42, 62; 2), (43, 47; 2), (43, 49; 2), (43, 54; 1), (43, 56; 1), (43, 62; 2), (44, 50; 1), (44, 55; 2), (44, 57; 1), (45, 54; 1), (46, 49; 2), (46, 54; 2), (47, 51; 1), (47, 56; 1), (47, 58; 2), (48, 56; 2), (49, 50; 2), (51, 55; 2), (53, 57; 2), (56, 58; 1), (56, 59; 2), (56, 61; 2), (56, 62; 1), (57, 60; 1), (57, 61; 2), (57, 62; 1), (61, 62; 2). Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $J_i = \varepsilon_i(\eta - r_i(\xi))$ for a unit ε_i of $\mathbb{Z}[\omega]$ and a polynomial r_i with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 4 gives $\eta = r_i(\xi)$ for the linear vertices. For a linear vertex i , we put $f_i(t) = F$ and $g_{ij}(t) = J_j$ evaluated at $\xi = t$, $\eta = r_i(t)$; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i , Table 5 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (4)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For every edge (i, j) with a linear endpoint, Table 6 gives the first linear endpoint $k \in \{i, j\}$ and the factorization of the norm of

the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $\omega^2 = -1 - \omega$.

5 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2. For $i \in \{4, 6, 7, 9, 11, 12, 13, 14, 15, 17, 20, 23, 28, 30, 31, 32, 36, 37, 39, 40, 44, 45, 47, 48, 49, 52, 53, 58, 61\}$, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{0, 1, 2, 3, 5, 21, 22, 24, 25, 42, 50, 51, 55, 56, 57, 59, 60\}$, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{8, 10, 29, 38, 43, 54\}$, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{16, 26, 46, 62\}$, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. For $i \in \{19, 34\}$, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $J_i \neq 0$. For $i \in \{41\}$, there is no residue pair (a, b) modulo 9 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{9}$, as one checks for the 9^2 residue pairs; hence $J_i \neq 0$. For $i \in \{18, 27, 33\}$, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $J_i \neq 0$. For $i \in \{35\}$, there is no residue pair (a, b) modulo 13 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{13}$, as one checks for the 13^2 residue pairs; hence $J_i \neq 0$. Also, all c_i are non-zero.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(1 + t)$ and the $\varphi_i(1 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-1 - t)$ and the $\varphi_i(-1 - t)$. Hence, for $x \geq 1$ and for $x \leq -1$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $x = 0$. Checking the residues of y , we find that for $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y . Hence (1) has no solution with $0 \leq x \leq 0$.

For a place v , put $B_v = \prod_{\{i, j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i < 0$ for $i \in \{1\}$. The other signs are given in Table 3 for $x \geq 1$ and for $x \leq -1$. In both cases, counting the edges whose two values are negative gives $B_\infty = 1$.

Step 3: the quadratic graphs at the odd primes outside S . Let $S = \{2, 3, 5, 7, 13, 19, 31\}$. It contains 2 and the primes dividing the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: cubic reciprocity. For every place v of K , put

$$\mathcal{B}_v = \sum_{(i,j)} w_{ij} \theta_v(J_i, J_j) + \sum_i \theta_v(J_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (5)$$

the first sum over the edges, and, for a rational prime p , put $\mathcal{B}_p = \sum_{v|p} \mathcal{B}_v$. Applying (H3) to every pair (J_i, J_j) with an edge, multiplied by w_{ij} , and to every pair (J_i, c_i) , and adding, we find that $\mathcal{B}_v = 0$ for all but finitely many v and $\sum_p \mathcal{B}_p = 0$, as the complex place contributes 0. By (H1) and the definition of cl_p , $\mathcal{B}_p = \sum_{(i,j)} w_{ij} \theta_p(\text{cl}_p(J_i), \text{cl}_p(J_j)) + \sum_i \theta_p(\text{cl}_p(J_i), \text{cl}_p(c_i))$, where θ_p is the form of Lemma 8 if $p = 3$.

Step 5: the cubic certificates at the primes outside S . The set S also contains 3 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides J_i and J_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $J_k = \varepsilon_k(\eta - r_k(\xi))$, we have $\eta \equiv r_k(t)$ modulo v , hence $f_k(t) \equiv F(x, y) = 0$ and $g_{kl}(t) \equiv J_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij} , a contradiction. By (H2), a term of (5) in which all arguments are units vanishes. Now let v divide J_i . Collecting the terms of (5) which contain J_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\theta_v(J_i, \Pi_i)$ with $\Pi_i = c_i \prod_j J_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in \mathbb{Z}[\omega]$. If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv J_j$ and $f_i(t) \equiv 0$ modulo v as above, and (4) gives the claim with $W = w_i(t)$. Hence the reduction of Π_i is a non-zero cube in k_v , $\chi(\Pi_i) = 0$, and $\theta_v(J_i, \Pi_i) = v(J_i)\chi(\Pi_i) = 0$ by (H2). Hence $\mathcal{B}_v = 0$, and $\mathcal{B}_p = 0$.

Step 6: the primes in S . Let $p \in S$, and suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Lemma 5 gives, for every vertex H_i of the quadratic graph, a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$, and Lemma 9 gives the analogous data for the values J_i of the cubic certificate. If the hypotheses of Lemma 6 hold for every quadratic graph and those of Lemma 10 hold for every cubic certificate, then B_p and $\mathcal{B}_p$ are determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until all these hypotheses hold, or until no solution remains. Table 7 lists the resulting terminal classes together with the joint value $(\mathcal{B}_p; B_p)$; every integer solution lies in one of them. The possible joint values are: for $p = 2$, $(0; 1)$; for $p = 3$, $(0; -1), (1; -1), (1; 1), (2; -1)$; for $p = 5$, $(0; 1)$; for $p = 7$, $(0; 1)$; for $p = 13$, $(0; 1)$; for $p = 19$, $(0; 1)$; for $p = 31$, $(0; 1)$.

Step 7: contradiction. By Steps 4 and 5, $\sum_{p \in S} \mathcal{B}_p = 0$. By Theorem 2(iv) and Step 3, $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 6, the pair $(\sum_{p \in S} \mathcal{B}_p; \prod_{p \in S} B_p)$ lies in the set consisting of the pairs $(0; -1)$, $(1; -1)$, $(1; 1)$, and $(2; -1)$, which does not contain $(0; 1)$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.
- [3] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The data of the identities (2).

$i = 0, m = 19, \text{neighbours } 1$	
W	$-16x^3 + 40x^2 - 16x + 3$
A	$-128x^2 + 640x + 64y - 704$
B	$64x^3y + 352x^3 - 64x^2y^2 - 320x^2y - 576x^2 + 288xy^2 + 480xy + 152x + 32y^3$ $-336y^2 - 200y + 316$

Table 2: The data of the identities (3).

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Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 1$ and for $x \leq -1$.

i	k_i	$\varphi_i(x)$	$x \geq 1$	$x \leq -1$
0	-2	$-16x^4 + 8x^3 - 12x^2 - 10x - 13$	+	+

Table 4: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle -y + 1, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, -x^2 \rangle$
1	$\langle -y, -x^2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 0, -x^2 \rangle$
2	$\langle x - y + 1, -x^2 + x \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, -x^2 + x \rangle$
3	$\langle -y + 1, x^2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 1, x^2 \rangle$
4	$\langle -y, x^2 - 1 \rangle$	$\langle -3, -3 \rangle$	$y = \langle 0, x^2 - 1 \rangle$
5	$\langle x - y + 1, x^2 \rangle$	$\langle -3, -3 \rangle$	$y = \langle x + 1, x^2 \rangle$
6	$\langle -x - y + 1, x^2 - x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, x^2 - x - 1 \rangle$
7	$\langle x - y + 1, x^2 + x + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x + 1, x^2 + x + 1 \rangle$
8	$\langle -x^2 - x - y + 1, -x^2 - x \rangle$	$\langle 1, -1 \rangle$	$y = \langle -x^2 - x + 1, -x^2 - x \rangle$
9	$\langle x^2 - y + 1, x^2 + 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2 + 1, x^2 + 1 \rangle$
10	$\langle x^2 - y - 1, x^2 + x \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2 - 1, x^2 + x \rangle$
11	$\langle x + 1, 1 \rangle$	$\langle -1, -1 \rangle$	$x = \langle -1, -1 \rangle$
12	$\langle x + 1, -1 \rangle$	$\langle -1, -1 \rangle$	$x = \langle -1, 1 \rangle$
13	$\langle x, 1 \rangle$	$\langle -1, -1 \rangle$	$x = \langle 0, -1 \rangle$
14	$\langle x, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, 1 \rangle$
15	$\langle x - 1, -1 \rangle$	$\langle 1, -1 \rangle$	$x = \langle 1, 1 \rangle$
16	$\langle x + y + 1, 0 \rangle$	$\langle -2, -4 \rangle$	$y = \langle -x - 1, 0 \rangle$
17	$\langle x + y, 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle -x, -1 \rangle$
18	$\langle x + y - 1, 0 \rangle$	$\langle -2, -4 \rangle$	$y = \langle -x + 1, 0 \rangle$
19	$\langle x - y - 1, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle x - 1, 0 \rangle$
20	$\langle x + 1, y - 1 \rangle$	$\langle 4, -4 \rangle$	$y = \langle x + 2, x + 1 \rangle$
21	$\langle x, y + 1 \rangle$	$\langle 0, -3 \rangle$	$y = \langle x - 1, x \rangle$
22	$\langle x, y \rangle$	$\langle 1, -1 \rangle$	$y = \langle x, x \rangle$
23	$\langle x - 1, -y \rangle$	$\langle 0, -4 \rangle$	$y = \langle -x + 1, -x + 1 \rangle$
24	$\langle x + y + 1, y + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -1, x \rangle$
25	$\langle x + y - 1, y - 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 1, x \rangle$
26	$\langle x + y^2 + 1, 0 \rangle$	$\langle -4, -4 \rangle$	$x = \langle -y^2 - 1, 0 \rangle$
27	$\langle -y + 1, 0 \rangle$	$\langle -2, -2 \rangle$	$y = \langle 1, 0 \rangle$
28	$\langle -y, -1 \rangle$	$\langle -2, -2 \rangle$	$y = \langle 0, -1 \rangle$
29	$\langle x^2 + x - y, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle x^2 + x, 0 \rangle$
30	$\langle x - y - 1, -1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x - 1, -1 \rangle$
31	$\langle 2x - y, x + 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle 2x, x + 1 \rangle$
32	$\langle -y - 1, -1 \rangle$	$\langle 0, -6 \rangle$	$y = \langle -1, -1 \rangle$
33	$\langle x - 2, 0 \rangle$	$\langle 2, 0 \rangle$	$x = \langle 2, 0 \rangle$
34	$\langle x^2 - y, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2, 0 \rangle$
35	$\langle x^2 - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - 1, 0 \rangle$

i	J_i	c_i	chart
36	$\langle x - y + 2, -y + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, -x - 1 \rangle$
37	$\langle x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 0 \rangle$
38	$\langle x + y, 0 \rangle$	$\langle -2, -1 \rangle$	$y = \langle -x, 0 \rangle$
39	$\langle x - y + 1, -y \rangle$	$\langle -1, 2 \rangle$	$y = \langle 0, -x - 1 \rangle$
40	$\langle x - y + 1, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, 1 \rangle$
41	$\langle -x^2 - y + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 + 1, 0 \rangle$
42	$\langle x^2 - y + 1, x^2 \rangle$	$\langle 0, -4 \rangle$	$y = \langle x^2 + 1, x^2 \rangle$
43	$\langle x^2 - x - y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - x, 0 \rangle$
44	$\langle -2x - y, -x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2x, -x + 1 \rangle$
45	$\langle x + 1, y + 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x, x + 1 \rangle$
46	$\langle -x^2 - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 1, 0 \rangle$
47	$\langle x - y - 2, x - 1 \rangle$	$\langle 0, -2 \rangle$	$y = \langle x - 2, x - 1 \rangle$
48	$\langle x + 2, 1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -2, -1 \rangle$
49	$\langle x - y - 2, -y - 1 \rangle$	$\langle 9, 24 \rangle$	$y = \langle -1, -x + 1 \rangle$
50	$\langle -x^2 - y - 1, -x^2 \rangle$	$\langle 2, -2 \rangle$	$y = \langle -x^2 - 1, -x^2 \rangle$
51	$\langle x + 2, 2 \rangle$	$\langle 4, 0 \rangle$	$x = \langle -2, -2 \rangle$
52	$\langle x^2 - y, x^2 - 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2, x^2 - 1 \rangle$
53	$\langle x + 1, y \rangle$	$\langle -4, -2 \rangle$	$y = \langle x + 1, x + 1 \rangle$
54	$\langle x - y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, 0 \rangle$
55	$\langle -x - y - 1, -x \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x - 1, -x \rangle$
56	$\langle x + y, y \rangle$	$\langle 0, -1 \rangle$	$y = \langle 0, x \rangle$
57	$\langle x, -y \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x, -x \rangle$
58	$\langle -x - y, -x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, -x - 1 \rangle$
59	$\langle x - y + 1, x \rangle$	$\langle 2, -2 \rangle$	$y = \langle x + 1, x \rangle$
60	$\langle -y + 1, -x \rangle$	$\langle 2, 0 \rangle$	$y = \langle 1, -x \rangle$
61	$\langle x + y^2 - y + 1, 0 \rangle$	$\langle 0, -1 \rangle$	$x = \langle -y^2 + y - 1, 0 \rangle$
62	$\langle x^2 - y + 1, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2 + 1, 0 \rangle$

Table 5: The data of the identities (4); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

$i = 0, D_i = 2, \rho_{ij}: 12:1, 15:2, 16:2, 19:1, 26:1, 28:1, 35:1, 39:2$
$w_i \quad \langle -6t^4 + 14t^3 + 4t^2 - 6t + 4, -4t^5 - 4t^4 + 12t^3 - 8t^2 + 2t + 4 \rangle$
$i = 1, D_i = 4, \rho_{ij}: 2:2, 6:2, 13:1, 17:1, 22:2, 24:2, 27:1, 28:2, 33:2, 39:2, 60:2$
$w_i \quad \langle 12t^5 - 12t^4 - 24t^3 + 12t^2 + 40t, 28t^5 - 44t^4 - 28t^3 + 8t^2 + 56t \rangle$
$i = 2, D_i = 64, \rho_{ij}: 1:1, 11:2, 18:1, 21:1, 22:2, 23:2, 32:1, 34:1, 40:1, 51:1$
$w_i \quad \langle 128t^5 + 448t^4 + 640t^3 + 256t^2 + 384t + 384, -128t^5 + 128t^4 + 128t^3 + 320t^2 + 512t + 128 \rangle$
$i = 3, D_i = 4, \rho_{ij}: 5:1, 9:1, 11:2, 13:1, 17:2, 26:1, 33:1, 37:1, 44:2, 46:1, 58:2$
$w_i \quad \langle 4t^5 - 16t^4 - 16t^3 - 4t^2 - 16t - 16, -12t^5 - 8t^4 - 16t^3 - 12t^2 - 16t - 16 \rangle$
$i = 4, D_i = 4, \rho_{ij}: 6:1, 9:2, 19:2, 25:2, 31:1, 35:1, 38:1, 39:2, 52:2$
$w_i \quad \langle 8t^5 - 20t^4 - 44t^3 - 44t^2 - 8t + 8, 16t^5 + 20t^4 - 16t^3 - 64t^2 - 28t - 8 \rangle$
$i = 5, D_i = 64, \rho_{ij}: 3:2, 8:1, 15:2, 17:2, 22:2, 27:1, 28:1, 33:1, 37:1$
$w_i \quad \langle 128t^4 + 256t^3 + 64t^2 - 128t - 256, 448t^4 - 256t^3 - 256t^2 - 64t - 128 \rangle$
$i = 6, D_i = 1024, \rho_{ij}: 1:1, 4:2, 7:1, 14:2, 22:2, 24:2, 28:1, 32:1, 42:2, 52:1$
$w_i \quad \langle 2048t^5 + 32768t^4 + 20480t^3 - 65536t^2 - 4096t + 28672, -6144t^5 + 14336t^4 + 65536t^3 - 4096t^2 - 43008t + 10240 \rangle$
$i = 7, D_i = 128, \rho_{ij}: 6:2, 12:1, 25:2, 40:1, 52:1, 53:2, 61:2$
$w_i \quad \langle 256t^5 + 256t^4 + 2048t^3 + 1792t^2 + 1024t, 1536t^5 + 1280t^4 + 1024t^3 + 512t^2 + 512t + 256 \rangle$
$i = 8, D_i = 256, \rho_{ij}: 5:2, 14:1, 17:1, 18:2, 22:2, 25:1, 33:1, 36:2, 38:2, 40:1, 48:1$
$w_i \quad \langle -7168t^5 - 22784t^4 + 2048t^3 + 14080t^2 + 4352t - 9728, 3328t^5 - 15616t^4 - 23552t^3 - 2560t^2 + 10240t - 4096 \rangle$
$i = 9, D_i = 4, \rho_{ij}: 3:2, 4:1, 11:2, 13:2, 20:2, 38:2, 55:1, 58:1$
$w_i \quad \langle 12t^5 + 20t^4 + 20t^3 - 4t^2 - 4t - 4, -4t^5 + 12t^4 + 24t^3 + 28t^2 - 4t - 8 \rangle$
$i = 10, D_i = 8, \rho_{ij}: 13:2, 29:1, 33:1, 37:1, 55:2, 56:1, 59:2$

w_i	$\langle 8t^4 - 16t^3 - 8t^2 + 8t + 16,$ $8t^4 - 8t^3 - 16t^2 - 8t + 16 \rangle$
$i = 11, D_i = 4, \rho_{ij}: 2:1, 3:1, 9:1, 23:1, 28:2, 29:2, 43:2, 49:1, 50:2, 59:2$	
w_i	$\langle 12t^2 - 16t + 32,$ $-4t^2 - 32t + 12 \rangle$
$i = 12, D_i = 4, \rho_{ij}: 0:2, 7:2, 19:1, 25:1, 29:1, 49:2$	
w_i	$\langle 12t^2 + 36t + 60,$ $24t + 32 \rangle$
$i = 13, D_i = 104, \rho_{ij}: 1:2, 3:2, 9:1, 10:1, 29:1, 31:1, 35:2, 43:2, 44:1, 46:2, 55:2, 61:1$	
w_i	$\langle -24t^2 + 384t + 1368,$ $1152t^2 + 600t - 1392 \rangle$
$i = 14, D_i = 8, \rho_{ij}: 6:1, 8:2, 16:1, 17:1, 20:1, 24:1, 27:1, 32:1, 36:2, 42:2, 43:2, 45:2, 47:2, 49:1, 53:1, 55:1$	
w_i	$\langle -56t^2 - 264t + 752,$ $-56t^2 - 144t + 120 \rangle$
$i = 15, D_i = 104, \rho_{ij}: 0:1, 5:1, 17:2, 19:2, 24:2, 25:1, 29:2, 39:2, 42:2, 45:1, 53:2$	
w_i	$\langle 272t^2 - 920t + 1656,$ $952t^2 + 368t + 1272 \rangle$
$i = 16, D_i = 16, \rho_{ij}: 0:1, 14:2, 21:2, 34:2, 35:2, 42:2, 44:2, 50:1$	
w_i	$\langle -442t^3 - 438t^2 + 66t + 332,$ $-95t^3 - 105t^2 - 21t + 34 \rangle$
$i = 17, D_i = 16, \rho_{ij}: 1:2, 3:1, 5:1, 8:2, 14:2, 15:1, 30:1, 33:2, 40:2, 41:2, 45:1, 56:2$	
w_i	$\langle -341t^3 + 812t^2 + 929t + 167,$ $-676t^3 - 203t^2 + 1033t + 289 \rangle$
$i = 18, D_i = 192, \rho_{ij}: 2:2, 8:1, 23:1, 26:2, 33:2, 34:1, 35:2, 40:2, 44:1, 50:2, 54:2, 58:2, 61:1, 62:1$	
w_i	$\langle -11353t^3 - 609t^2 - 4383t - 3130,$ $-5261t^3 - 6069t^2 + 4149t - 9794 \rangle$
$i = 19, D_i = 2, \rho_{ij}: 0:2, 4:1, 12:2, 15:1, 26:1, 33:1, 37:1, 38:1, 41:2$	
w_i	$\langle 24t^3 - 16t + 32,$ $5t^3 + t^2 - 3t + 22 \rangle$
$i = 20, D_i = 2, \rho_{ij}: 9:1, 14:2, 23:1, 38:2, 48:1, 52:2$	
w_i	$\langle 10t^3 + 6t^2 - 2,$ $8t^3 - 6t^2 - 12t - 10 \rangle$
$i = 21, D_i = 4, \rho_{ij}: 2:2, 16:1, 23:1, 25:1, 26:1, 33:1, 35:1, 50:2$	
w_i	$\langle 13t^3 - 29t^2 - 31t + 2,$ $29t^3 - 16t^2 - 11t - 2 \rangle$
$i = 22, D_i = 4, \rho_{ij}: 1:1, 2:1, 5:1, 6:1, 8:1, 23:2, 37:2, 51:2, 54:2$	
w_i	$\langle -22t^3 + 4t^2 + 6t + 28,$ $-14t^3 - 16t^2 - 24t - 4 \rangle$
$i = 23, D_i = 2, \rho_{ij}: 2:1, 11:2, 18:2, 20:2, 21:2, 22:1, 27:2, 30:1, 41:1, 52:1, 53:1$	
w_i	$\langle -78t^3 + 66t^2 - 24t - 34,$ $-126t^3 + 174t^2 - 122t - 16 \rangle$
$i = 24, D_i = 4, \rho_{ij}: 1:1, 6:1, 14:2, 15:1, 27:2, 33:2, 36:2, 37:2, 38:1, 39:1, 44:1, 45:1, 49:1, 53:2, 57:2$	
w_i	$\langle -100t^3 + 142t^2 - 84t + 432,$ $-346t^3 + 324t^2 - 6t + 156 \rangle$
$i = 25, D_i = 8, \rho_{ij}: 4:1, 7:1, 8:2, 12:2, 15:2, 21:2, 31:1, 37:2, 39:1, 49:1, 52:1, 53:2$	
w_i	$\langle -42t^3 - 25t^2 + 62t,$ $-43t^3 + 34t^2 + 7t - 22 \rangle$
$i = 26, D_i = 2, \rho_{ij}: 0:2, 3:2, 18:1, 19:2, 21:2, 27:2, 33:2, 34:1, 53:2$	
w_i	$\langle -6t^7 - 2t^6 - 17t^5 - 6t^4 - 8t^3 - 18t^2 - t - 6,$ $-6t^7 - 2t^6 - 17t^5 - 6t^4 - 8t^3 - 18t^2 - t - 6 \rangle$
$i = 27, D_i = 2, \rho_{ij}: 1:2, 5:2, 14:2, 23:1, 24:1, 26:1, 40:2$	
w_i	$\langle -2t^3 + t + 2,$ $-2t^3 + 4t^2 \rangle$
$i = 28, D_i = 28, \rho_{ij}: 0:2, 1:1, 5:2, 6:2, 11:1, 37:2, 39:1$	
w_i	$\langle -28t^3 - 40t^2 - 26t + 28,$ $40t^3 - 14t^2 - 68t - 20 \rangle$
$i = 29, D_i = 32768, \rho_{ij}: 10:2, 11:1, 12:2, 13:2, 15:1, 30:1, 39:2, 41:2, 42:1, 51:2, 54:1, 56:2, 57:2, 59:1, 60:1, 61:1$	
w_i	$\langle -2097152t^5 + 29229056t^4 + 13041664t^3 - 4980736t^2 - 10354688t + 15073280,$ $-20381696t^5 - 16252928t^4 + 327680t^3 + 7274496t^2 - 8126464t - 3670016 \rangle$
$i = 30, D_i = 32, \rho_{ij}: 17:2, 23:2, 29:2, 35:2, 37:2, 41:1, 47:1, 48:1, 50:2, 52:1, 55:1$	
w_i	$\langle 70t^3 - 524t^2 + 1074t - 350,$ $-534t^3 + 406t^2 + 600t - 872 \rangle$
$i = 31, D_i = 4, \rho_{ij}: 4:2, 13:2, 25:2, 32:2, 46:2$	
w_i	$\langle 4t^2 - 2t + 6,$ $-6t^3 + 2t^2 + 2t \rangle$
$i = 32, D_i = 62, \rho_{ij}: 2:2, 6:2, 14:2, 31:1, 39:1$	
w_i	$\langle -146t^3 + 86t^2 + 250t + 480,$ $-70t^3 - 104t^2 - 112t + 90 \rangle$

$i = 33, D_i = 32, \rho_{ij}:$	1:1, 3:2, 5:2, 8:2, 10:2, 17:1, 18:1, 19:2, 21:2, 24:1, 26:1, 34:1, 36:2, 39:2, 46:1, 55:1, 56:2, 57:2, 59:2, 60:1, 62:2
w_i	$\langle 9653760t^2 - 26707968t + 98505216, -17326080t^2 + 57950208t - 144138240 \rangle$
$i = 34, D_i = 4, \rho_{ij}:$	2:2, 16:1, 18:2, 26:2, 33:2, 37:2, 40:2, 56:1, 57:1
w_i	$\langle 92t^5 + 44t^4 - 80t^3 - 4t^2 + 108t + 56, 0 \rangle$
$i = 35, D_i = 4, \rho_{ij}:$	0:2, 4:2, 13:1, 16:1, 18:1, 21:2, 30:1, 50:1, 52:2, 55:2
w_i	$\langle 8t^4 + 24t^3 + 12t^2 + 16, -24t^5 + 4t^4 + 24t^3 - 12t + 8 \rangle$
$i = 36, D_i = 2, \rho_{ij}:$	8:1, 14:1, 24:1, 33:1, 44:1, 52:2, 57:1
w_i	$\langle -8t^2 - 14t - 6, 4t^3 - 4t^2 - 16t - 14 \rangle$
$i = 37, D_i = 8, \rho_{ij}:$	3:2, 5:2, 10:2, 19:2, 22:1, 24:1, 25:1, 28:1, 30:1, 34:1, 42:1, 44:2, 53:1
w_i	$\langle -16t^2 - 80t + 72, -32t^2 - 16t + 72 \rangle$
$i = 38, D_i = 4, \rho_{ij}:$	4:2, 8:1, 9:1, 19:2, 20:1, 24:2, 42:1, 52:1
w_i	$\langle 4t^3 + 4, -4t^3 - 12t^2 - 12t - 4 \rangle$
$i = 39, D_i = 8, \rho_{ij}:$	0:1, 1:1, 4:1, 15:1, 24:2, 25:2, 28:2, 29:1, 32:2, 33:1, 52:2, 55:2, 56:1
w_i	$\langle 213t^3 + 285t^2 + 216t + 45, 344t^3 + 438t^2 + 219t - 55 \rangle$
$i = 40, D_i = 4, \rho_{ij}:$	2:2, 7:2, 8:2, 17:1, 18:1, 27:1, 34:1, 53:1
w_i	$\langle 4t^3 - 29t^2 - 7t + 3, 19t^3 - 4t^2 - 25t + 9 \rangle$
$i = 41, D_i = 4, \rho_{ij}:$	17:1, 19:1, 23:2, 29:1, 30:2, 51:2, 59:1, 61:1
w_i	$\langle -16t^5 - 24t^4 + 12t^3 - 4t - 8, -20t^5 - 48t^4 + 20t^3 + 20t^2 - 28t - 24 \rangle$
$i = 42, D_i = 4, \rho_{ij}:$	6:1, 14:1, 15:1, 16:1, 29:2, 37:2, 38:2, 45:1, 48:1, 50:1, 53:1, 61:1, 62:2
w_i	$\langle 8t^5 - 32t^4 + 72t^3 + 80t^2 + 104t + 48, 48t^5 + 64t^4 + 80t^3 + 80t^2 + 64t + 64 \rangle$
$i = 43, D_i = 32, \rho_{ij}:$	11:1, 13:1, 14:1, 47:2, 49:2, 54:1, 56:1, 62:2
w_i	$\langle 0, 32t^5 - 64t^4 + 32t^3 - 64t^2 + 32t - 64 \rangle$
$i = 44, D_i = 4, \rho_{ij}:$	3:1, 13:2, 16:1, 18:2, 24:2, 36:2, 37:1, 50:1, 55:2, 57:1
w_i	$\langle 20t^3 - 24t^2 + 12t + 8, 16t^3 - 16t^2 + 4t \rangle$
$i = 45, D_i = 2, \rho_{ij}:$	14:1, 15:2, 17:2, 24:2, 42:2, 54:1
w_i	$\langle 3t^3 - t^2 - 6t + 1, -2t^3 + t^2 - 4t - 1 \rangle$
$i = 46, D_i = 4, \rho_{ij}:$	3:2, 13:1, 31:1, 33:2, 49:2, 54:2
w_i	$\langle -4t^5 + 4t^4 + 4t^3 + 8t^2, 4t^3 - 4t^2 - 4t - 8 \rangle$
$i = 47, D_i = 2, \rho_{ij}:$	14:1, 30:2, 43:1, 51:1, 56:1, 58:2
w_i	$\langle 4t^2 + 4t + 4, -4t^3 + 4 \rangle$
$i = 48, D_i = 4, \rho_{ij}:$	8:2, 20:2, 30:2, 42:2, 56:2
w_i	$\langle 4t^2 - 8t, 8t - 12 \rangle$
$i = 49, D_i = 4, \rho_{ij}:$	11:2, 12:1, 14:2, 24:2, 25:2, 43:1, 46:1, 50:2
w_i	$\langle -76t^3 - 140t^2 + 256t - 164, -128t^3 + 140t^2 + 20t - 64 \rangle$
$i = 50, D_i = 2, \rho_{ij}:$	11:1, 16:2, 18:1, 21:1, 30:1, 35:2, 42:2, 44:2, 49:1
w_i	$\langle -64t^5 - 64t^4 + 16t^3 - 28t^2 - 48t - 16, -20t^5 - 56t^4 + 44t^3 + 16t^2 - 48t - 32 \rangle$
$i = 51, D_i = 2, \rho_{ij}:$	2:2, 22:1, 29:1, 41:1, 47:2, 55:2
w_i	$\langle 4t^2 + 12t + 40, -16t + 32 \rangle$
$i = 52, D_i = 4, \rho_{ij}:$	4:1, 6:2, 7:2, 20:1, 23:2, 25:2, 30:2, 35:1, 36:1, 38:2, 39:1
w_i	$\langle -52t^5 - 108t^4 + 48t^3 + 144t^2 - 32t - 48, 24t^5 + 88t^4 + 80t^3 + 172t^2 + 40t - 8 \rangle$
$i = 53, D_i = 4, \rho_{ij}:$	7:1, 14:2, 15:1, 23:2, 24:1, 25:1, 26:1, 37:2, 40:2, 42:2, 57:2
w_i	$\langle -58t^3 - 86t^2 - 78t - 20, 52t^3 + 56t^2 + 6t - 34 \rangle$
$i = 54, D_i = 4, \rho_{ij}:$	18:1, 22:1, 29:2, 43:2, 45:2, 46:1
w_i	$\langle 6t^3 + 8t^2 + 4, 6t^3 + 8t^2 + 4 \rangle$
$i = 55, D_i = 16, \rho_{ij}:$	9:2, 10:1, 13:1, 14:2, 30:2, 33:2, 35:1, 39:1, 44:1, 51:1

w_i	$\langle 52t^3 + 74t^2 + 124t + 32, 50t^3 + 86t^2 + 102t + 28 \rangle$
$i = 56, D_i = 608, \rho_{ij}:$	$10:2, 17:1, 29:1, 33:1, 34:2, 39:2, 43:2, 47:2, 48:1, 58:1, 59:2, 61:2, 62:1$
w_i	$\langle -7084t^3 - 6166t^2 + 2559t + 4080, 511t^3 + 2388t^2 + 2170t + 2030 \rangle$
$i = 57, D_i = 4, \rho_{ij}:$	$24:1, 29:1, 33:1, 34:2, 36:2, 44:2, 53:1, 60:1, 61:2, 62:1$
w_i	$\langle 12t^3 + 38t^2 + 49t, 51t^3 + 50t^2 - 29t - 50 \rangle$
$i = 58, D_i = 2, \rho_{ij}:$	$3:1, 9:2, 18:1, 47:1, 56:2$
w_i	$\langle -2t^3 - 2t^2 + 2, -4t^3 - 6t^2 - 8t - 2 \rangle$
$i = 59, D_i = 4, \rho_{ij}:$	$10:1, 11:1, 29:2, 33:1, 41:2, 56:1$
w_i	$\langle 8t^3 + 2t^2 + 8t + 8, 2t^3 - 6t^2 + 2t + 4 \rangle$
$i = 60, D_i = 2, \rho_{ij}:$	$1:1, 29:2, 33:2, 57:2$
w_i	$\langle -2t^3 - 6t - 4, -2t^3 + 2t^2 - 2t - 4 \rangle$
$i = 61, D_i = 8, \rho_{ij}:$	$7:1, 13:2, 18:2, 29:2, 41:2, 42:2, 56:1, 57:1, 62:2$
w_i	$\langle 8t^7 - 36t^6 + 100t^5 - 158t^4 + 162t^3 - 84t^2 + 10t + 12, -8t^7 + 38t^6 - 82t^5 + 116t^4 - 122t^3 + 116t^2 - 84t + 36 \rangle$
$i = 62, D_i = 4, \rho_{ij}:$	$18:2, 33:1, 42:1, 43:1, 56:2, 57:2, 61:1$
w_i	$\langle -4t^5 - 24t^4 - 16t^2 - 16t - 16, -4t^5 - 24t^4 - 16t^2 - 16t - 16 \rangle$

Table 6: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(0, 12)	0	2^4	(18, 23)	18	5^2
(0, 15)	0	$2^2 \cdot 3$	(18, 26)	18	2^8
(0, 16)	0	$2^6 \cdot 3^4$	(18, 33)	18	$2^4 \cdot 3^4$
(0, 19)	0	2^8	(18, 34)	18	$2^4 \cdot 19^2$
(0, 26)	0	$2^{10} \cdot 3$	(18, 35)	18	$2^6 \cdot 5^2 \cdot 7^2$
(0, 28)	0	2^2	(18, 40)	18	$2^2 \cdot 7 \cdot 19$
(0, 35)	0	$2^2 \cdot 3$	(18, 44)	18	$7 \cdot 19$
(0, 39)	0	3^3	(18, 50)	18	$2^6 \cdot 19$
(1, 2)	1	7	(18, 54)	18	$2^4 \cdot 3^4$
(1, 6)	1	$7 \cdot 13^2 \cdot 19$	(18, 58)	18	2^2
(1, 13)	1	3	(18, 61)	18	1
(1, 17)	1	$7 \cdot 19$	(18, 62)	18	$2^2 \cdot 3^4$
(1, 22)	1	$2^2 \cdot 7$	(19, 26)	19	2^{14}
(1, 24)	1	$13 \cdot 19$	(19, 33)	19	$2^4 \cdot 3^4$
(1, 27)	1	13	(19, 37)	19	3^2
(1, 28)	1	$3 \cdot 13$	(19, 38)	19	$2^8 \cdot 3^2$
(1, 33)	1	$2^4 \cdot 3 \cdot 19$	(19, 41)	19	$2^6 \cdot 5^2$
(1, 39)	1	19	(20, 23)	20	$2^2 \cdot 3^3 \cdot 7$
(1, 60)	1	$3^2 \cdot 7$	(20, 38)	20	3^6
(2, 11)	2	7^2	(20, 48)	20	$2^4 \cdot 13$
(2, 18)	2	$2^2 \cdot 7 \cdot 19$	(20, 52)	20	$3^4 \cdot 7 \cdot 13$
(2, 21)	2	$2^2 \cdot 7^4$	(21, 23)	21	$2^2 \cdot 7^2 \cdot 13$
(2, 22)	2	$3 \cdot 7$	(21, 25)	21	$2^6 \cdot 3 \cdot 7$
(2, 23)	2	$3 \cdot 7^4$	(21, 26)	21	$2^6 \cdot 3 \cdot 7^2$
(2, 32)	2	$3^3 \cdot 7^2$	(21, 33)	21	$2^6 \cdot 3 \cdot 7$
(2, 34)	2	$3 \cdot 19^2$	(21, 35)	21	$2^2 \cdot 3 \cdot 7$
(2, 40)	2	$3^3 \cdot 7$	(21, 50)	21	$2^2 \cdot 3 \cdot 13$
(2, 51)	2	$2^2 \cdot 3^4 \cdot 7$	(22, 23)	22	$2^6 \cdot 3 \cdot 7$
(3, 5)	3	2^2	(22, 37)	22	$3 \cdot 7$
(3, 9)	3	$2^2 \cdot 13$	(22, 51)	22	$2^6 \cdot 3 \cdot 7$
(3, 11)	3	2^2	(22, 54)	22	2^2
(3, 13)	3	$3 \cdot 19$	(23, 27)	23	3

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(3, 17)	3	19	(23, 30)	23	$7 \cdot 13^2$
(3, 26)	3	$2^{10} \cdot 7$	(23, 41)	23	$5^2 \cdot 7^2$
(3, 33)	3	$2^4 \cdot 3 \cdot 7 \cdot 19$	(23, 52)	23	$3^2 \cdot 7^2$
(3, 37)	3	3	(23, 53)	23	2^8
(3, 44)	3	19	(24, 27)	24	$2^2 \cdot 3 \cdot 7 \cdot 13$
(3, 46)	3	$2^2 \cdot 7$	(24, 33)	24	$2^6 \cdot 3 \cdot 13$
(3, 58)	3	13	(24, 36)	24	$2^6 \cdot 3^2 \cdot 13$
(4, 6)	4	$2^2 \cdot 3$	(24, 37)	24	$3 \cdot 7$
(4, 9)	4	19	(24, 38)	24	3^2
(4, 19)	4	$2^8 \cdot 5^2$	(24, 39)	24	$2^2 \cdot 7 \cdot 19$
(4, 25)	4	3	(24, 44)	24	$2^4 \cdot 3 \cdot 7^2$
(4, 31)	4	1	(24, 45)	24	7
(4, 35)	4	$3^8 \cdot 5^2$	(24, 49)	24	$2^4 \cdot 3 \cdot 7^2$
(4, 38)	4	19	(24, 53)	24	7^3
(4, 39)	4	3^2	(24, 57)	24	$7^2 \cdot 13$
(4, 52)	4	1	(25, 31)	25	$2^2 \cdot 7$
(5, 8)	5	$2^2 \cdot 3^6 \cdot 7^2$	(25, 37)	25	$3 \cdot 7$
(5, 15)	5	3	(25, 39)	25	$2^2 \cdot 3^2 \cdot 7$
(5, 17)	5	$5^2 \cdot 7^2$	(25, 49)	25	$2^4 \cdot 5^2 \cdot 7$
(5, 22)	5	$3^2 \cdot 7$	(25, 52)	25	$2^4 \cdot 7$
(5, 27)	5	$2^2 \cdot 3$	(25, 53)	25	3^3
(5, 28)	5	1	(26, 27)	26	2^{10}
(5, 33)	5	$2^4 \cdot 3 \cdot 5^2$	(26, 33)	26	$2^{12} \cdot 3^2 \cdot 7^2$
(5, 37)	5	$3 \cdot 7$	(26, 34)	26	1
(6, 7)	6	2^{14}	(26, 53)	26	$3^2 \cdot 7$
(6, 14)	6	$2^8 \cdot 7$	(27, 40)	27	3
(6, 22)	6	$2^4 \cdot 7^2$	(28, 37)	28	3
(6, 24)	6	$2^8 \cdot 7 \cdot 19$	(28, 39)	28	1
(6, 28)	6	$2^2 \cdot 13$	(29, 30)	29	$3^2 \cdot 7^2$
(6, 32)	6	$2^4 \cdot 7^3$	(29, 39)	29	$3^4 \cdot 7$
(6, 42)	6	$2^2 \cdot 3^2 \cdot 7$	(29, 41)	29	$3^4 \cdot 7^4$
(6, 52)	6	$2^4 \cdot 7$	(29, 42)	29	$3^2 \cdot 19$
(7, 12)	7	$2^2 \cdot 19^2$	(29, 51)	29	$2^2 \cdot 3^2 \cdot 7$
(7, 25)	7	$2^4 \cdot 7 \cdot 19$	(29, 54)	29	2^4
(7, 40)	7	$3 \cdot 7$	(29, 56)	29	$2^2 \cdot 7 \cdot 19$
(7, 52)	7	$2^4 \cdot 3^2$	(29, 57)	29	$2^2 \cdot 7 \cdot 19$
(7, 53)	7	1	(29, 59)	29	$2^4 \cdot 7$
(7, 61)	7	$3^3 \cdot 19$	(29, 60)	29	$2^4 \cdot 7$
(8, 14)	8	7^2	(29, 61)	29	$3^2 \cdot 7^2 \cdot 19^2$
(8, 17)	8	$7 \cdot 19$	(30, 35)	30	$3 \cdot 7 \cdot 19$
(8, 18)	8	2^4	(30, 37)	30	$3 \cdot 13$
(8, 22)	8	$2^8 \cdot 3^2 \cdot 7$	(30, 41)	30	$3^4 \cdot 7^2$
(8, 25)	8	2^2	(30, 47)	30	$7 \cdot 13$
(8, 33)	8	$2^8 \cdot 3^2 \cdot 19$	(30, 48)	30	13
(8, 36)	8	$2^2 \cdot 3^6$	(30, 50)	30	$3^2 \cdot 13$
(8, 38)	8	3^4	(30, 52)	30	$7 \cdot 13 \cdot 19$
(8, 40)	8	$3^4 \cdot 19$	(30, 55)	30	$7^2 \cdot 13$
(8, 48)	8	1	(31, 32)	31	$3^4 \cdot 7$
(9, 11)	9	2^4	(31, 46)	31	$3^2 \cdot 7$
(9, 13)	9	3	(32, 39)	32	7
(9, 20)	9	$2^4 \cdot 3^3$	(33, 34)	33	$2^{10} \cdot 3^2$
(9, 38)	9	$3^4 \cdot 19$	(33, 36)	33	$3^2 \cdot 7 \cdot 13$
(9, 55)	9	3	(33, 39)	33	$3^2 \cdot 7$
(9, 58)	9	$2^4 \cdot 3 \cdot 13$	(33, 46)	33	$2^4 \cdot 3^2 \cdot 7^2$
(10, 13)	10	3	(33, 55)	33	$2^6 \cdot 3 \cdot 13$
(10, 29)	10	$3^2 \cdot 7$	(33, 56)	33	$2^2 \cdot 3 \cdot 13^2$
(10, 33)	10	$2^6 \cdot 3^3 \cdot 13$	(33, 57)	33	$2^2 \cdot 3 \cdot 13^2$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(10, 37)	10	3^2	(33, 59)	33	$2^8 \cdot 3$
(10, 55)	10	$2^2 \cdot 13^2$	(33, 60)	33	$2^8 \cdot 3$
(10, 56)	10	13	(33, 62)	33	$2^4 \cdot 3^2 \cdot 13^2$
(10, 59)	10	$2^4 \cdot 7$	(34, 37)	34	3^2
(11, 23)	11	$7 \cdot 13$	(34, 40)	34	$3 \cdot 19^2$
(11, 28)	11	2^2	(34, 56)	34	2^6
(11, 29)	11	7	(34, 57)	34	2^6
(11, 43)	11	$7 \cdot 13$	(35, 50)	35	$2^4 \cdot 3^6$
(11, 49)	11	$2^2 \cdot 7^2$	(35, 52)	35	$7^2 \cdot 19$
(11, 50)	11	7	(35, 55)	35	$2^2 \cdot 7$
(11, 59)	11	7^2	(36, 44)	36	$2^4 \cdot 3^3 \cdot 7$
(12, 19)	12	2^8	(36, 52)	36	$3^4 \cdot 7$
(12, 25)	12	$5^2 \cdot 19$	(36, 57)	36	$2^4 \cdot 13$
(12, 29)	12	$7 \cdot 19$	(37, 42)	37	$3 \cdot 7$
(12, 49)	12	$2^2 \cdot 5^2 \cdot 7$	(37, 44)	37	$3 \cdot 13$
(13, 29)	13	$3^2 \cdot 7$	(37, 53)	37	3^2
(13, 31)	13	3	(38, 42)	38	1
(13, 35)	13	$3 \cdot 7$	(38, 52)	38	3^4
(13, 43)	13	3	(39, 52)	39	7
(13, 44)	13	$3 \cdot 7 \cdot 19$	(39, 55)	39	7^3
(13, 46)	13	3^2	(39, 56)	39	$2^6 \cdot 7$
(13, 55)	13	$3 \cdot 7$	(40, 53)	40	1
(13, 61)	13	3^4	(41, 51)	41	$2^2 \cdot 3^4 \cdot 7^2$
(14, 16)	14	2^4	(41, 59)	41	$2^2 \cdot 7^2$
(14, 17)	14	$7 \cdot 13$	(41, 61)	41	$3^2 \cdot 7^4$
(14, 20)	14	2^2	(42, 45)	42	$2^2 \cdot 7^2$
(14, 24)	14	7	(42, 48)	42	2^4
(14, 27)	14	7	(42, 50)	42	$2^{12} \cdot 3 \cdot 7$
(14, 32)	14	2^4	(42, 53)	42	3^3
(14, 36)	14	7^2	(42, 61)	42	$3 \cdot 19$
(14, 42)	14	2^2	(42, 62)	42	2^4
(14, 43)	14	$7 \cdot 13$	(43, 47)	43	$7^2 \cdot 13$
(14, 45)	14	$2^2 \cdot 13$	(43, 49)	43	$7^2 \cdot 13$
(14, 47)	14	$2^2 \cdot 7^2$	(43, 54)	43	$2^4 \cdot 3^2 \cdot 7^2$
(14, 49)	14	$7 \cdot 13$	(43, 56)	43	$2^2 \cdot 3$
(14, 53)	14	7	(43, 62)	43	3^4
(14, 55)	14	7	(44, 50)	44	$7 \cdot 13^2 \cdot 19$
(15, 17)	15	$3 \cdot 13$	(44, 55)	44	3^2
(15, 19)	15	$2^2 \cdot 3$	(44, 57)	44	7
(15, 24)	15	$3 \cdot 7$	(45, 54)	45	3^2
(15, 25)	15	3	(46, 49)	46	$3^2 \cdot 7^2$
(15, 29)	15	$3^2 \cdot 7$	(46, 54)	46	7^2
(15, 39)	15	3^3	(47, 51)	47	$3 \cdot 7^2$
(15, 42)	15	3	(47, 56)	47	$2^6 \cdot 7$
(15, 45)	15	$3^2 \cdot 13$	(47, 58)	47	$2^{10} \cdot 3$
(15, 53)	15	$3^2 \cdot 7$	(48, 56)	48	$2^4 \cdot 13$
(16, 21)	16	$2^2 \cdot 3^4$	(49, 50)	49	7
(16, 34)	16	2^8	(51, 55)	51	$2^2 \cdot 3^2 \cdot 7$
(16, 35)	16	$2^2 \cdot 3^2$	(53, 57)	53	$2^6 \cdot 7$
(16, 42)	16	$2^6 \cdot 3^4$	(56, 58)	56	2^6
(16, 44)	16	1	(56, 59)	56	3
(16, 50)	16	$2^4 \cdot 3^2$	(56, 61)	56	19^2
(17, 30)	17	$2^4 \cdot 3 \cdot 7^2$	(56, 62)	56	$3 \cdot 13 \cdot 19$
(17, 33)	17	$3 \cdot 5^2 \cdot 19$	(57, 60)	57	3
(17, 40)	17	$2^4 \cdot 7 \cdot 19$	(57, 61)	57	19^2
(17, 41)	17	$3^2 \cdot 7^2$	(57, 62)	57	$3 \cdot 13 \cdot 19$
(17, 45)	17	13^3	(61, 62)	61	19^2

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
$(17, 56)$	17	7			

Table 7: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 6, grouped by p, k and the joint value $(\mathcal{B}_p; B_p)$.

p	k	value	classes (a, b)
2	2	(0; 1)	(0, 2)
2	3	(0; 1)	(2, 4), (6, 0), (6, 3), (6, 5), (6, 7)
2	6	(0; 1)	(14, 9), (14, 41), (30, 1), (30, 33), (46, 25), (46, 57), (62, 17), (62, 49)
3	3	(0; -1)	(5, 5), (14, 23), (23, 14)
3	3	(1; -1)	(8, 2), (8, 21), (17, 12), (17, 20), (26, 3), (26, 11)
3	3	(1; 1)	(2, 10), (5, 7), (8, 4), (11, 1), (14, 25), (17, 22), (20, 19), (23, 16), (26, 13)
3	3	(2; -1)	(2, 8), (11, 26), (20, 17)
3	4	(0; -1)	(2, 36), (11, 54), (20, 72), (29, 9), (38, 27), (47, 45), (56, 63), (65, 0), (74, 18)
3	4	(2; -1)	(5, 15), (14, 33), (23, 51), (32, 69), (41, 6), (50, 24), (59, 42), (68, 60), (77, 78)
5	1	(0; 1)	(1, 1), (1, 4), (2, 3)
5	2	(0; 1)	(1, 5), (6, 0), (11, 20), (16, 15), (21, 10)
7	2	(0; 1)	(3, 19), (4, 28), (4, 36), (5, 31), (6, 11), (10, 19), (11, 29), (12, 31), (13, 11), (18, 7), (18, 22), (19, 31), (20, 11), (24, 19), (25, 15), (26, 31), (27, 11), (31, 19), (32, 8), (32, 35), (33, 31), (34, 11), (37, 23), (38, 19), (39, 0), (39, 1), (41, 11), (44, 44), (45, 19), (46, 14), (46, 41), (46, 43), (47, 31), (48, 11)
7	3	(0; 1)	(2, 16), (4, 34), (9, 135), (11, 76), (11, 140), (18, 118), (23, 79), (25, 160), (25, 266), (30, 247), (32, 202), (39, 244), (40, 31), (51, 163), (53, 328), (58, 282), (60, 27), (60, 238), (66, 19), (67, 69), (72, 226), (74, 21), (74, 111), (79, 51), (81, 153), (88, 195), (89, 31), (100, 310), (102, 279), (107, 86), (109, 321), (109, 336), (115, 19), (116, 20), (121, 30), (123, 62), (123, 119), (128, 198), (130, 104), (137, 146), (138, 31), (149, 114), (151, 230), (156, 233), (158, 91), (158, 272), (163, 254), (164, 19), (165, 314), (170, 177), (172, 13), (172, 217), (177, 2), (179, 55), (186, 97), (187, 31), (198, 261), (200, 181), (205, 37), (207, 189), (207, 223), (213, 19), (214, 265), (219, 324), (221, 307), (221, 315), (226, 149), (228, 6), (235, 48), (236, 31), (247, 65), (249, 132), (254, 184), (256, 174), (256, 287), (262, 19), (263, 216), (268, 128), (270, 70), (270, 258), (275, 296), (277, 300), (284, 342), (285, 31), (296, 212), (298, 83), (303, 331), (305, 42), (305, 125), (310, 9), (311, 19), (312, 167), (317, 275), (319, 168), (319, 209), (324, 100), (326, 251), (333, 293), (334, 31)
7	4	(0; 1)	(16, 1528), (17, 362), (65, 1332), (114, 1136), (212, 744), (261, 548), (359, 156), (360, 362), (408, 2361), (457, 2165), (555, 1773), (604, 1577), (702, 1185), (703, 362), (751, 989), (800, 793), (898, 401), (947, 205), (1045, 2214), (1046, 362), (1094, 2018), (1143, 1822), (1241, 1430), (1290, 1234), (1388, 842), (1389, 362), (1437, 646), (1486, 450), (1584, 58), (1633, 2263), (1731, 1871), (1732, 362), (1780, 1675), (1829, 1479), (1927, 1087), (1976, 891), (2074, 499), (2075, 362), (2123, 303), (2172, 107), (2270, 2116), (2319, 1920)
13	1	(0; 1)	(0, 3), (0, 7), (3, 10), (4, 10), (6, 0), (6, 12), (10, 4)

p	k	value	classes (a, b)
13	2	(0; 1)	(1, 17), (3, 162), (4, 45), (6, 53), (9, 164), (11, 24), (12, 112), (14, 108), (15, 44), (16, 123), (17, 19), (19, 53), (22, 151), (24, 37), (25, 47), (27, 30), (28, 44), (29, 84), (32, 53), (35, 138), (37, 50), (38, 151), (41, 44), (42, 45), (43, 136), (45, 53), (48, 125), (50, 63), (51, 86), (53, 43), (54, 44), (55, 6), (56, 110), (58, 53), (61, 112), (63, 76), (64, 21), (66, 134), (67, 44), (68, 136), (69, 84), (71, 53), (74, 99), (76, 89), (77, 125), (79, 56), (80, 44), (81, 97), (82, 58), (84, 53), (87, 86), (89, 102), (90, 60), (92, 147), (93, 44), (94, 58), (95, 32), (97, 53), (100, 73), (102, 115), (103, 164), (105, 69), (106, 44), (107, 19), (108, 6), (110, 53), (113, 60), (115, 128), (116, 99), (118, 160), (119, 44), (120, 149), (121, 149), (123, 53), (126, 47), (128, 141), (129, 34), (131, 82), (132, 44), (133, 110), (134, 123), (136, 53), (139, 34), (141, 154), (142, 138), (144, 4), (146, 71), (147, 97), (149, 53), (152, 21), (154, 167), (155, 73), (157, 95), (158, 44), (159, 32), (160, 71), (162, 53), (165, 8), (167, 11), (168, 8)
13	3	(0; 1)	(2, 1565), (30, 2021), (40, 459), (145, 1565), (171, 1565), (199, 1683), (209, 1642), (314, 1565), (340, 1565), (368, 1345), (378, 628), (483, 1565), (509, 1565), (537, 1007), (547, 1811), (652, 1565), (678, 1565), (706, 669), (716, 797), (821, 1565), (847, 1565), (875, 331), (885, 1980), (990, 1565), (1016, 1565), (1044, 2190), (1054, 966), (1159, 1565), (1185, 1565), (1213, 1852), (1223, 2149), (1328, 1565), (1354, 1565), (1382, 1514), (1392, 1135), (1497, 1565), (1523, 1565), (1551, 1176), (1561, 121), (1666, 1565), (1692, 1565), (1720, 838), (1730, 1304), (1835, 1565), (1861, 1565), (1889, 500), (1899, 290), (2004, 1565), (2030, 1565), (2058, 162), (2068, 1473), (2173, 1565)
19	1	(0; 1)	(0, 9), (1, 16), (8, 5), (8, 12), (11, 4), (12, 8), (15, 4), (17, 12), (18, 15)

p	k	value	classes (a, b)
19	2	(0; 1)	(2, 143), (4, 339), (5, 224), (6, 270), (8, 306), (10, 339), (16, 179), (17, 24), (17, 173), (21, 29), (23, 339), (24, 357), (25, 270), (27, 154), (29, 282), (35, 293), (36, 249), (36, 328), (40, 276), (42, 339), (43, 129), (44, 270), (46, 2), (48, 225), (54, 46), (55, 271), (55, 325), (59, 162), (61, 339), (62, 262), (63, 270), (65, 211), (67, 168), (73, 160), (74, 40), (74, 214), (78, 48), (80, 339), (81, 34), (82, 270), (84, 59), (86, 111), (92, 274), (93, 116), (93, 157), (97, 295), (99, 339), (100, 167), (101, 270), (103, 268), (105, 54), (111, 27), (112, 100), (112, 192), (116, 181), (118, 339), (119, 300), (120, 270), (122, 116), (124, 358), (130, 141), (131, 43), (131, 268), (135, 67), (137, 339), (138, 72), (139, 270), (141, 325), (143, 301), (149, 255), (150, 344), (150, 347), (154, 314), (156, 339), (157, 205), (158, 270), (160, 173), (162, 244), (168, 8), (169, 59), (169, 290), (173, 200), (175, 339), (176, 338), (177, 270), (179, 21), (181, 187), (187, 122), (188, 135), (188, 233), (192, 86), (194, 339), (195, 110), (196, 270), (198, 230), (200, 130), (206, 236), (207, 176), (207, 211), (211, 333), (213, 339), (214, 243), (215, 270), (217, 78), (219, 73), (225, 350), (226, 119), (226, 287), (230, 219), (232, 339), (233, 15), (234, 270), (236, 287), (238, 16), (244, 103), (245, 2), (245, 62), (249, 105), (251, 339), (252, 148), (253, 270), (255, 135), (257, 320), (263, 217), (264, 5), (264, 78), (268, 352), (271, 281), (272, 270), (274, 344), (276, 263), (282, 331), (283, 154), (283, 309), (287, 238), (289, 339), (290, 53), (291, 270), (293, 192), (295, 206), (301, 84), (302, 230), (302, 252), (306, 124), (308, 339), (309, 186), (310, 270), (312, 40), (314, 149), (320, 198), (321, 195), (321, 306), (325, 10), (327, 339), (328, 319), (329, 270), (331, 249), (333, 92), (339, 312), (340, 21), (340, 138), (344, 257), (346, 339), (347, 91), (348, 270), (350, 97), (352, 35), (358, 65), (359, 81), (359, 97)
19	3	(0; 1)	(270, 2866), (631, 2866), (992, 2866), (1353, 2866), (1714, 2866), (2075, 2866), (2436, 2866), (2797, 2866), (3158, 2866), (3519, 2866), (3880, 2866), (4241, 2866), (4602, 2866), (4963, 2866), (5324, 2866), (5685, 2866), (6046, 2866), (6407, 2866), (6768, 2866)
31	1	(0; 1)	(0, 4), (0, 10), (0, 17), (1, 7), (2, 7), (3, 16), (5, 13), (6, 26), (7, 29), (8, 20), (9, 12), (10, 13), (11, 0), (11, 1), (11, 30), (12, 23), (13, 5), (17, 4), (17, 10), (17, 17), (22, 4), (22, 10), (22, 17), (23, 4), (23, 10), (23, 17), (24, 8), (26, 3), (26, 6), (26, 22), (27, 18), (28, 29)